ALGEBRA 2 · QUADRATIC FUNCTIONS & COMPLEX NUMBERS

Discriminant & Roots Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Discriminant & Roots Explorer** on Mr. K's Classroom: Algebra 2 → Quadratic Functions & Complex Numbers → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If a parabola never touches the x-axis, does the equation have real solutions?

2 Explore use the tool and record what you see

Discriminant and roots

Equation	$b^2 - 4ac$	Number and type of roots	Roots
$x^2 - 2x - 3 = 0$			
$x^2 - 6x + 9 = 0$			
$x^2 + 4 = 0$			
$x^2 - 2x + 2 = 0$			
$x^2 + 2x + 5 = 0$			
$2x^2 + 3x - 2 = 0$			

3 Write the rule fill in each blank from what you observed

1. $b^2 - 4ac > 0$: _____. $b^2 - 4ac = 0$: _____. $b^2 - 4ac < 0$: _____.

2. Complex roots of a quadratic with real coefficients come in _____ pairs, $a + bi$ and $a - bi$.

3. On the complex plane, conjugate roots are reflections over the _____ axis.

4. If the parabola opens up and its vertex is above the x-axis, the roots are _____.

4 Check no tool: show your work

1. Find the discriminant of $3x^2 - 4x + 5$ and describe the roots.

2. Solve $x^2 + 9 = 0$.

3. Solve $x^2 - 4x + 13 = 0$.

4. For what c does $x^2 + 6x + c = 0$ have exactly one real root?

5 Reflect

What does the discriminant tell you about the graph?